Five Pascal Carriers. One Quotient.

Independent review of a finite typed graph with 106 vertices, 180 directed recurrence edges and candidate full D5 symmetry.

106	180	D5	177 / 177
quotient vertices	directed edges	candidate full typed group	internal verifier checks

WHY A HUMAN REVIEW

Mathematics can now be explored, challenged and documented with AI at unusual speed. What AI cannot provide is the independent human judgement that decides whether a definition is sound, a proof closes, a nearby construction has been missed, or a contribution is worth carrying forward. SCC-003-R1 is offered here to request that return.

AI systems assisted with exploration, criticism and documentation. Finite computational claims were also checked by deterministic programs. Neither layer is treated as independent scientific validation. The mathematical object, the claims and responsibility for their presentation remain with the author.

OPEN REVIEW DRAFT / NOT PEER REVIEWED / NO NOVELTY OR PRIORITY CLAIM / INDEPENDENT HUMAN REVIEW REQUESTED

Claims presented for review

The review is deliberately finite. A valid return may accept, revise or reject any claim, identify controlling prior art, or conclude that the review is incomplete.

ID	Claim	Current state
SCC003-C1	Five labelled depth-six Pascal carriers glued by the declared full side-chain equivalence produce open counts 28 49 70 91 112	INTERNAL PROVED
SCC003-C2	Cyclic closure identifies six additional non-apex pairs and produces 106 quotient vertices	INTERNAL PROVED
SCC003-C3	The quotient has 180 unique directed recurrence edges	INTERNAL VERIFIED
SCC003-C4	The quotient is connected and retains facewise Pascal recurrence	INTERNAL VERIFIED
SCC003-C5	The declared rotations and reflections act as typed automorphisms forming D5	INTERNAL VERIFIED
SCC003-C6	The full typed automorphism group is exactly D5	INTERNAL VERIFIED
SCC003-N1	No exact or controlling predecessor was located in the bounded search	COVERAGE LIMIT
SCC003-N2	The construction is externally novel or first	NOT ESTABLISHED
SCC003-P1	The construction supports a physical resonance or natural mechanism	OUT OF SCOPE

Abstract

We define a finite weighted complex by cyclically gluing five copies of the depth-six Pascal carrier along complete radial side chains. The successive open quotients have `28, 49, 70, 91, 112` vertices. Closing the remaining terminal seam identifies six pairs of non-apex vertices and produces a connected quotient with `106` vertices and `180` directed recurrence edges. The binomial weights and Pascal recurrence are retained facewise. Rotation of the five faces and reflection across one face axis descend to typed automorphisms, and the resulting typed automorphism group is exactly the dihedral group `D_5` of order `10`. We give a deterministic reference object, an independent union-find verifier and a bounded comparison with nearby Pascal-triangle graph constructions. The note claims neither worldwide novelty nor a physical interpretation; independent mathematical and literature review remain open.

1. Scope and claim boundary

The object of this note is one finite combinatorial construction. It is not a statement about the whole NEXAH system and does not depend on the A0 book, interactive explorer, metric pentagon, three-dimensional lift, root-system view, modular-shadow language or pentagon/hexagon visual analogies.

The permitted contribution statement is:

> *A cyclic five-face quotient of depth-six binomially weighted Pascal*
>
> *carriers with explicit* `D_5` *automorphisms and open-to-closed node sequence*
>
> `28,49,70,91,112 -> 106`.

This is a candidate contribution statement. It is not a priority, novelty, significance or publication claim.

2. The carrier

Let

```
T_6 = {(n,k) in Z^2 : 0 <= k <= n <= 6}.
```

The depth-six Pascal carrier is the directed weighted graph

```
P_6 = (T_6, E_6, w_6),

E_6 = {
  ((n,k),(n+1,k)),
  ((n,k),(n+1,k+1))
  : 0 <= k <= n < 6
},

w_6(n,k) = binom(n,k).
```

It has 28 vertices and 42 directed edge incidences. Its two radial sides are

```
L = {(n,0):0<=n<=6},
R = {(n,n):0<=n<=6}.
```

We retain two typed layers: the triangular macro-face used for gluing and the directed recurrence graph inside that face.

3. Five copies and the seam relation

Index five copies by `C_5 = Z/5Z`. A raw vertex is `(f,n,k)` with `f in C_5` and `(n,k) in T_6`. Identify all five apices, and for every face index `f` and level `1<=n<=6` impose

```
(f,n,n) ~ (f+1,n,0).
```

The exclusion of `n=0` avoids recounting the already common apex. The equivalence relation generated by these pairs has:

- one apex class of size five;
- thirty non-apex seam classes of size two; and
- seventy-five singleton interior classes.

The closed object is not an anonymous graph. It is the quotient together with its weights, directed adjacency and five retained face-incidence maps.

4. Open construction and cyclic closure

For `1<=j<=5`, let `K_open(j)` contain faces `0,...,j-1`, with common apex and only the consecutive seam identifications. The two terminal sides remain unpaired. Each new face shares exactly the apex and six non-apex side vertices. It therefore contributes

```
28 - 7 = 21
```

new vertices. Hence

```
|V(K_open(j))| = 28 + 21(j-1) = 21j + 7,
```

which yields

```
28, 49, 70, 91, 112.
```

Closing the terminal seam of `K_open(5)` pairs its twelve terminal non-apex vertices level by level. The six pairs are disjoint and generate no further identifications, so

```
|V(K_closed)| = 112 - 6 = 106.
```

The corresponding six duplicate side edges reduce the directed quotient-edge count from `186` to `180`.

5. Weights, recurrence and connectivity

Define the quotient weight by

```
w([f,n,k]) = binom(n,k).
```

This is well-defined because every nontrivial non-apex class identifies two side vertices of weight one; the apex class also has constant weight one.

For every retained face incidence, Pascal's identity remains

```
binom(n,k) = binom(n-1,k-1) + binom(n-1,k),
```

with zero extension outside `0<=k<=n`. The assertion is deliberately facewise. Summing every incoming edge of the simple quotient graph would erase face incidence and can double-count a glued seam.

Every `P_6` is connected when directions are ignored, and all face apices are identified. Thus every open complex is connected. A quotient of a connected graph is connected, so `K_closed` is connected.

6. Dihedral action

On raw positions define

```
r(f,n,k) = (f+1,n,k),
m(f,n,k) = (-f,n,n-k),
```

with face indices modulo five. Both maps preserve the seam relation and descend to the quotient. They preserve level, weight, directed recurrence edges and macro-face incidence, and satisfy

```
r^5 = e,
m^2 = e,
m r m = r^-1.
```

The ten maps `{r^a, m r^a : 0<=a<5}` therefore act as typed automorphisms. Conversely, every typed automorphism induces an automorphism of the retained five-face adjacency cycle. This action is injective because fixing a face and its two seams fixes every internal Pascal position. Since `Aut(C_5)=D_5`, the typed automorphism group is exactly `D_5`.

This statement concerns the weighted, directed, face-incidence object. It is not an assertion that a graph6 string alone certifies the full automorphism group of every untyped relabeling.

7. Canonical reference object

The source package exports:

- `106` named quotient vertices;
- `180` directed quotient edges;
- all weights and seam classes;
- `140` raw face positions and `210` face-edge incidences;
- a directed adjacency matrix;
- a declared-order graph6 serialization of the underlying simple graph;
- the ten typed `D_5` permutations;
- expected metrics and a SHA-256 manifest.

A separate verifier reconstructs the quotient with an independent union-find implementation and checks the finite counts, recurrences, adjacency, hashes and dihedral relations. The current verification result is `177/177` checks passed.

Canonical package: SCC-003 reference object.

8. Nearby constructions

The bounded prior-art search located two especially relevant neighbors:

1. The Hilton-Pedersen Pascal-Hexagon line extends ordinary binomial values

through six triangular sextants and studies cross-sector patterns. It is the strongest multi-sector Pascal baseline found, but it does not define five finite copies, complete side-chain gluing or the `112 -> 106` quotient.

2. Quint's infinite Pascal graph is the closest literal graph-joining

baseline found: two parity-derived triangular graphs are joined at their apex vertices. It is infinite, unweighted in the present sense, uses two copies and has no complete side-chain quotient.

The audit also compared binary Pascal/Steinhaus symmetry, Pascal divisibility networks, Pascal hyper-pyramids and generic quotient-polynomial graphs. No inspected source matched all controlling features. Coverage remained incomplete for exhaustive MathSciNet/zbMATH exports and has not been repeated by an independent reviewer.

The controlled search outcome is therefore:

```
CLOSE_BASELINE_FOUND_NO_EQUIVALENCE_ESTABLISHED
SEARCH_INCOMPLETE_ACCESS_OR_COVERAGE_LIMIT
```

Full comparison: bounded prior-art audit.

9. Contextual representations outside the theorem

Separate bounded tests realize the same finite carrier in a regular metric pentagon, as one hemisphere of a pentagonal bipyramid, and beside the standard root completion `H_2 direct-sum A_1`. A separate `C_5/C_6` control embeds both cycles into a common `C_30` phase cover after direct identification fails.

These are typed representations and comparators. They are not needed for the definition or proofs above and do not establish a universal Pascal solid, physical resonance, crystallographic Lie system, canonical role for `110/111/112`, or an identity between pentagonal and hexagonal carriers.

10. Limitations and next gate

This note fixes the object and compacts the existing proof and reproducibility record. It does not:

- generalize from five faces or depth six;
- prove worldwide novelty or priority;
- establish physical meaning or empirical utility;
- activate a Science Lab study;
- authorize external submission or publication.

The next gate is an independent mathematical and literature review. Useful review questions are:

1. Is the seam equivalence relation unambiguous and complete?
2. Do the open-count, closure, recurrence and connectivity proofs cover all

quotient edge cases?

3. Is the typed-automorphism exactness argument sufficient?
4. Does an equivalent or immediately specializing construction already exist?
5. Is the reference object enough for independent reconstruction?

No reviewer has been selected or contacted.

Companion-work boundary

The later Pascal-Rosetta C12 relation, D30 phase lift, five-register view and H3/E8 non-identity audit are intentionally **not** added to this theorem. They use the same frozen raw incidences but ask a different question about typed addresses, quotient return and representation loss. Mixing them into the SCC-003 abstract would weaken the paper's single-object claim boundary.

They are documented separately in A typed D30 phase lift for a five-register Pascal carrier. The A0 Findings and Paper Workspace keeps that companion draft, the closed Return findings and later archaeological outlines distinct. Neither document changes Admission, evidence or publication status.

References

- P. Hilton, J. Pedersen and W. Rosenthal, *Pascalian triangles and extensions

to hexagons*, Quaestiones Mathematicae 13 (1990), 395-416. Accessible construction: <https://www.mi.sanu.ac.rs/vismath/hpe/hpe1.htm>.

- J.-F. Quint, *Harmonic analysis on the Pascal graph*, Journal of Functional

Analysis 256 (2009), 3409-3460. <https://doi.org/10.1016/j.jfa.2009.01.011>.

- J. M. Brunat and M. Maureso, *Rotational and dihedral symmetries in

Steinhaus and Pascal binary triangles* (2011).

- A. Barbé, *Symmetric patterns in the cellular automaton that generates

Pascal's triangle modulo 2*, Discrete Applied Mathematics 105 (2000). <https://doi.org/10.1016/S0166-218X(00)00211-0>.

- E. Solares-Hernandez et al., *Divisibility Patterns within Pascal

Divisibility Networks*, Mathematics 8 (2020), 254. <https://doi.org/10.3390/math8020254>.

- M. E. Horn, *Pascal Pyramids, Pascal Hyper-Pyramids and a Bilateral

Multinomial Theorem* (2003). <https://arxiv.org/abs/math/0311035>.

Source hierarchy

The concise definitions above are a paper view. If wording differs, the source-bound formal specification, proposition package and deterministic data artifact control the mathematical object:

- formal specification;
- propositions and proofs;

- canonical reference package;
- SCC-003 candidate assessment.

Reviewer brief

Requested expertise

Primary reviewer: finite graph theory, algebraic combinatorics, enumerative combinatorics or simplicial/quotient constructions. Literature reviewer: mathematical information specialist or mathematician able to search MathSciNet, zbMATH, Crossref, arXiv and citation networks.

Object

Let `P_6={(r,k):0<=k<=r<=6}` with weight `C(r,k)` and directed Pascal recurrence edges. Take five labelled copies. Consecutive copies share a complete seven-node side chain under the declared seam equivalence, all apices are identified, and the last side chain is closed to the first. The quotient is typed by weights, directed recurrence, face incidence and seam provenance.

The submitted propositions establish:

1. open counts `28,49,70,91,112`;
2. cyclic closure identifies six additional non-apex pairs and gives 106

quotient vertices;

3. the quotient has 180 unique directed recurrence edges;
4. it is connected and preserves the facewise Pascal recurrence;
5. its declared rotations and reflections form a typed `D5` action;
6. the full typed automorphism group is exactly the ten declared elements.

Why review is requested

The authors have a formal specification, compact proofs, a deterministic reference export and a separately implemented local verifier. A bounded prior-art audit located nearby Pascal graphs, multi-sector Pascal extensions, binary Pascal symmetries, Pascal networks and generalized Pascal pyramids, but no exact predecessor. Database coverage and specialist review remain incomplete.

What is not claimed

- no universal Pascal algebra or general `m,d` theorem;
- no physical, quantum, harmonic, elemental or astronomical mechanism;
- no novelty inference from the number 106;
- no claim that metric, 3D, root-system, D30 or quaternion representations are

part of the narrow theorem;

- no worldwide priority claim.

Requested mode

Please review the formal object before inspecting visual or contextual materials. A prior-art objection should identify an explicit mapping from a published construction to the five controlling features in the claim table; terminological resemblance alone is insufficient.

If a proof fails, report the earliest failing definition or inference. If a predecessor is found, cite the exact theorem/construction and classify it as equivalent, immediately specializing to SCC-003, or merely analogous.

Return contract

- Name, field, relevant qualification and conflict-of-interest statement.
- Exact packet-manifest SHA-256 and replay environment.
- The earliest failing definition or inference if a proof does not close.
- A precise structural mapping for any proposed predecessor.
- One bounded disposition: accept correctness, revise, reject, prior art found, no equivalence found within scope, or review incomplete.

Packet ID: SCC003-ERP-2026-10-13
Manifest SHA-256: 73b0322764620a5faef7a1b92f7cdb552ea39d09e3a79386b6454377679996f2
Identity reused: SCC-003 / A0OE:PASCAL_PENTAGON_D5_TEST